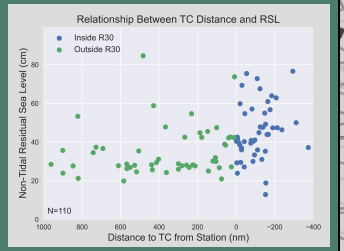
Investigating the Relationship between Historical Storm Surge and Distance to Storm Center in the Philippines

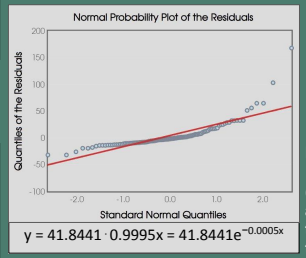
High Meadows Environmental Institute

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ENV/CEE/GEO 480

Motivation

Throughout the West Pacific, the Philippines is one of the most frequently hit countries by typhoons. This results in it being especially important to understand the exposure and vulnerabilities to typhoon hazards. With the mix of low-lying coastal areas and various bays, storm surge is especially dangerous in these coastal areas, and understanding the hazard intensity is vital. Following new research investigating historical analysis of storm surge throughout the Philippines, this study aims to better understand the connection between tropical cyclone path and storm surge. While previous studies have explored the effects of storm surge through models and simulations, this study aims to build off of the approach from Ada Ilagan et al. (2026) and capture storm surge pathing through real-world, physical data. In order to do this, it will combine the International Best Track Archive for Climate Stewardship (IBTrACS) database and the real-world tide gauges to explore and identify how important typhoon proximity impacts gauge heights. The aim of this study is to review our current understanding of typhoon pathing, and confirm whether or not it aligns with real-world Observed data.





$y = 41.8441 \cdot 0.9995x = 41.8441e^{-0.0005x}$

Data

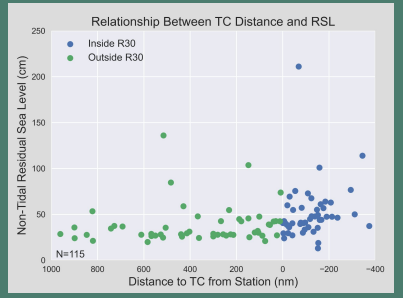
For the 133 typhoons identified from 1974 to 2024 from Ada Ilagan et al. (2026), the maximum storm surge height across all stations was recorded. At that time step, the distance to the center of the storm was also collected. Then from the IBTrACS database, the storms were then collected and wind speeds, and the radius of >30 knot wind speeds were collected. Eleven storms did not have sufficient wind data at that time step and were consequently removed. Eight storms were above 1000 mb at the timestep as well, not meeting the tropical cyclone threshold, and were consequently removed. Of the 115 remaining storms, they were then plotted.

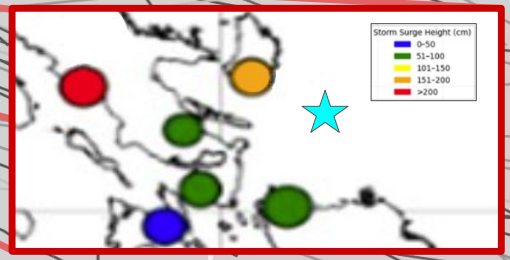
Key Motivating Questions

1. To what extent is there a relationship between the size of and distance to a TC and storm surge?
2. How much does the TC wind speeds impact local storm surge levels?
3. Does observed storm surge data align with what we would expect based on our current model understandings?

Methods

For the 115 typhoons selected, R_Overlap was calculated by subtracting the radius of the storm from the distance to the storm center. Therefore, negative R_Overlap indicates that the location was inside the >30 knot ring. This was plotted against the Non-Tidal Residual Sea Level (RSL), the storm surge. After removing major outliers, and stratifying the data, a pattern where decreasing the R_Overlap resulted in the RSL increased exponentially. Specifically, once it reached the >30 knot ring, it started increasing rapidly. However, an exponential regression was unable to capture this, despite the Q-Q residual graph indicating some slight exponential curve.





Discussion

In brief, this study finds that water gauge data reinforces the effects of typhoon pathing over the Philippines. While there are some outliers in the data, like at the Pasacao station, there seems to be a minor indication of an intensification of storm surge depending on whether or not >30 knot winds go overhead. These findings are important to reiterate how even these relatively low speeds are at a significant threshold where hazards become exponentially more extreme.

[a] https://coast.noaa.gov/hurricanes
[b] Anjela Ada Ilagan, Olivia C. Cabrera, Marcelino Q. Villafuerte II et al.,
"Investigating Historical Storm Surge Occurrences in the Philippines from Tide Gauge Observations", 09 January 2026
[c] NOAA's International Best Track Archive for Climate Stewardship (IBTrACS) data, accessed on April 18th, 2026